

Progress Update: CMAB Evaluation Complete & iCMAB Testing Underway

Piter Garcia (RIT Student) <pzg8794@rit.edu>
Reply-To: pzg8794@rit.edu
To: Daniel Krutz <dxkvse@rit.edu>

Fri, Oct 10, 2025 at 8:46 AM

Sounds good

On Fri, Oct 10, 2025 at 8:45AM Daniel Krutz <dxkvse@rit.edu> wrote:
You can join and just stay muted.

On Oct 10, 2025, at 8:44AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Should I come back in?

On Tue, Oct 7, 2025 at 3:45 PM Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:
Yes, that works!

I will send an invite.

Best,
Piter

On Tue, Oct 7, 2025 at 12:47 PM Daniel Krutz <dxkvse@rit.edu> wrote:
Hi Piter, thanks for the updates. Is Friday at 830 good for you for a zoom?

Thanks
-Dan

On Oct 6, 2025, at 3:03AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Dan,

Status Update:

iCMAB/CMAB Testing: All baseline testing complete. Currently debugging minor OOP issues in the iCMAB framework to ensure proper integration with our evaluation system. Expect initial hybrid model results by Tuesday.

This Week's Deliverables:

- Complete hybrid model evaluations
- Finalize documentation
- Begin preliminary work on quantum routing optimizations

Proposed Meeting: Friday morning works best—gives me time to wrap up testing and prepare comprehensive results, including early insights on qubit allocation adjustments.

Does Friday morning work for you?

Best,
Piter

On Sat, Oct 4, 2025 at 11:13AM Daniel Krutz <dxkvse@rit.edu> wrote:
Hey Piter, no need for the apologies.

Your plan looks good to me. When do you want to touch base next? About when will you be ready with more results?

Thanks
-Dan

On Oct 3, 2025, at 6:45AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Dan,

My apologies for not emailing earlier—I was swamped with coursework and preparing for a teaching lesson. I'll be on the lookout for your email this morning in case you'd like to meet, but below is my progress update.

Quick update on the quantum MAB evaluation framework:

Phase 1 (Complete): Contextual MAB Baseline Testing

- Completed evaluation across 3, 5, 8, and 10 run configurations
- **Key finding:**
Contextual models significantly outperform pure neural approaches in stochastic environments:
 - CPursuit:
94.4%
avg efficiency (variance 10.9)
 - CEpsilonGreedy:
92.4%
(variance 3.1)
 - CThompsonSampling: 88.7% (variance 3.4)
- These results show a 15-20% performance advantage over baseline neural models
- Statistical validation confirms consistency across all run configurations ($p < 0.001$)

Phase 2 (In Progress): iCMAB Evaluation

- Currently running iCMAB versions of top-performing CMAB algorithms
- Using the same robust testing methodology to identify the strongest candidates
- Goal: Select baselines, test competitors, and identify high contenders for hybrid development

Parallel Work: Next-Gen Hybrid Development

- Constructing a hybrid model using top contenders (CPursuit, CEpsilonGreedy, CThompsonSampling) to replace the EXP3 component
- Based on current results, achieving performance gains over ExpNeuralUCB is highly feasible, especially in stochastic scenarios
- This parallel approach will save significant time once the iCMAB evaluation is completed

Target: Finalize iCMAB testing and identify the optimal hybrid contender demonstrating measurable improvement over ExpNeuralUCB by the end of the weekend. Next week: Wrap experiments/results and pivot to investigating advanced circuit allocation strategies and quantum routing optimizations.

Notebooks and detailed reports available for review:

- [3 Runs Configuration](#)
- [5 Runs Configuration](#)
- [8 Runs Configuration](#)
- [10 Runs Configuration](#)

Best,
Piter

